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A Supervised Neural Network Machine Learning Routing Algorithm to Improve the Lifetime of Heterogeneous Wireless Sensor Networks

Ahmed Gamal[1] · Amr ElSaadany[2] · SHAWKAT K. GUIRGUIS[3]

Abstract

Internet of Things has many applications requiring the use of wireless communications networks. It utilizes data collection from sensor nodes connected to Wireless Sensor Networks. As such the wireless sensor networks is considered an important key for data transmission between sensor node and the gateways which are connected to the internet. Of the main concerns is the lifetime of the network which is affected by the battery power of the sensor nodes. It is noteworthy that transmission energy that dominates overall energy consumption is proportional to the distance between the transmitter and receiver. Thus, there is a need to send data from source node to a destination node in the most efficient way when it comes to battery level. Although there are many algorithms that tried to perform energy efficient routing, we will propose an intelligent algorithm to further improve this routing problem. In this paper we proposed an AI algorithm to enhance the lifetime of wireless sensor networks. We show that our algorithm improves the lifetime of the network by up to 75%, depending on the traffic rate, over existing algorithms.

Keywords Wireless Sensor Networks, Neural Networks, Machine Learning, Routing Algorithm, Shortest Path, Lifetime

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1 Introduction

In recent years, the use of the Internet in our daily life has taken a big leap from being a luxury to a necessity. Internet of Things (IoT) is the next revolutionary technology that will transform the Internet into a fully integrated future internet. The basic idea of IoT is the pervasive presence of a variety of things or objects which are able to interact with each other to achieve common goals. US National Intelligence Council NIC foresees that" by 2025 Internet nodes may reside in everyday things - food packages, furniture, paper documents, and more"[1]. The IoT transforms these objects from being traditional to smart by exploiting its underlying technologies such as ubiquitous and pervasive computing, embedded devices, communication technologies, sensor networks, Internet protocols, and applications. Figure 1 shows the wireless sensor network as part of the technologies supporting the Internet of Things [2].

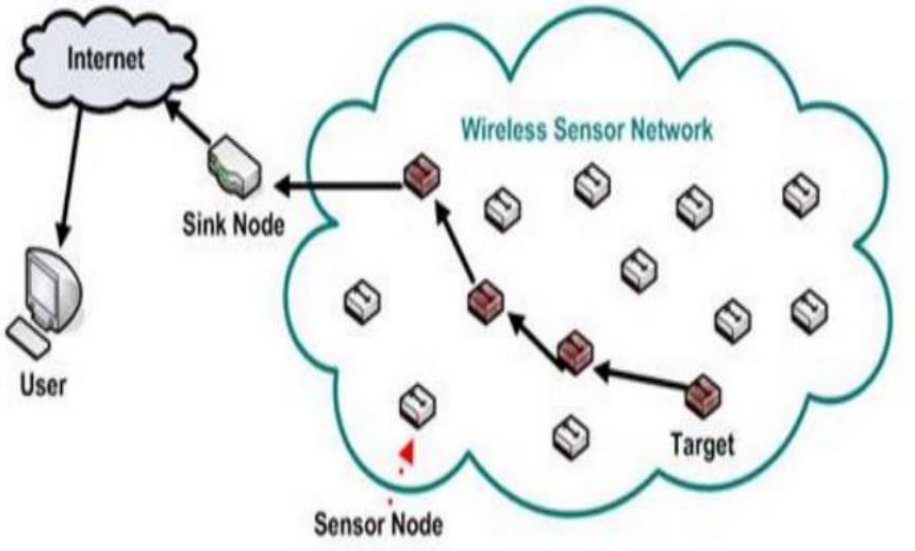


Fig. 1. WSN Architecture on IoT Environment

The Wireless sensor networks (WSNs) are most important area of research due to the realization of their ability in myriad applications. WSNs is including varies applications as military surveillance, facility monitoring, target detection, and health care applications. Moreover, WSNs are considered as one of the key enablers to the IoT era [3, 4].

A WSN comprises of a number of spatially distributed sensor nodes which have low power of battery energy [5, 6], which are used to monitor environmental conditions for various metrics, such metric varies from temperature, pressure, noise, vibration, [7] as well as detecting object motion or existence of pollutants, Typically, WSNs are densely deployed in hazardous places where battery recharge or replacement is nearly impossible [8]. Also human monitoring scheme is highly risky, There are many typical issues such as power constraints, limited computing capacity, open environment; radio connectivity makes the sensor nodes faulty many times. Once the network is established, nodes keep on sensing the information and the battery power increases exponentially. Whenever the nodes detect any event, they send the information to the other nodes or to the base station.

There is always a need for algorithms and methods that reduce energy consumption and increase the lifetime for WSNs as the nodes have limited battery source. It is noteworthy that transmission energy that dominates overall energy consumption is proportional to the distance between the transmitter and receiver [8]. Thus, there is a need to send data from

source node to a destination node in the most efficient way when it comes to battery level. There is quite a large number of papers that address the battery consumption and there relation to the routing algorithms [9]. Although there are many algorithms that tried to perform energy efficient routing, we will propose an intelligent algorithm to further improve this routing problem.
AI algorithms are powerful methods to solve this issue and obtain the optimum solution [10].In this paper, we focus on proposing a novel algorithm to improve lifetime WSNs using neural networks to obtain on optimal route that have low link cost and save the residual energy of nodes. A main design objective of the proposed algorithm is energy efficiency, lifetime maximization and decrease data rate for WSNs.

2 Literature Survey

In this section we provide a summary of the most related work done in this area. The work ranges from algorithms that use fuzzy logic and neural network to heuristic methods. In doing so, we set the stage to introduce our new proposed algorithm which is based on artificial intelligent techniques.

2.1. Distributed Energy-aware Fuzzy Logic based routing algorithm (DEFL)

The Distributed Energy-aware Fuzzy Logic based routing algorithm (DEFL) was proposed in [11], the authors proposed DEFL based on shortest path routing strategy with minimum cost because the most important factors in energy consumption is transmission power because it depends on the distance between node i (source) and node j (destination) which energy required to transmit one unit of information from a node i to node j is given by equation 1:

$$E_t\,(ij) = \beta_1 + \beta_2\, d_{ij}^4 \qquad (1)$$

Where d_{ij} represents the distance between the two nodes, β_1 = 50nJ/bit is the energy consumed to run the transmitter circuitry, and β_2 = 100pJ/bit/m4 is the energy consumed at the transmitter amplifier. The energy required to run the receiver circuitry is assumed to be constant and specified as β_3 = 150nJ/bit.
Furthermore, there are heterogeneous nodes in WSNs i.e. nodes with different initial battery capacity, and operate under variable traffic conditions, so there is a need to design an optimum cost function which will achieve network life time maximization in networks
In the DEFL algorithm, both the energy level and the energy consumption rate are used as inputs to a fuzzy logic mapping mechanism into a single cost value. This energy related cost values are assigned to the network links in order to use the shortest path to find the minimum total path cost routes.
The DEFL algorithm is based on two fuzzy logic systems that use triangle membership function. It uses five fuzzy rules for fuzzy system1 (FS1) and twenty-five rules for fuzzy system2 (FS2) and five fuzzy sets (very high, high, medium, low, very low) for each input and output variable of both fuzzy systems. The domain of all input variables has been adjusted such that the universal of discourse is between 0 and 1; this allows the fuzzy sets to be applicable for any network configuration.
The metrics used in this algorithm for applying to FS1 and FS2 are three inputs: -

- Transmission Energy (TE), these metric measures the amount of energy needed to perform a transmission.
- Remaining Energy (RE), this metric measures the instantaneous battery level of a node.

- Energy Drain Rate (DR), this metric measures the rate at which the energy is consumed over a time period.

These three inputs are applied to two fuzzy systems and triangle membership functions used for all input and output variables of both fuzzy systems.
There are two fuzzy logics FS1 and FS2 are based mapping systems produce relay probabilities which are then converted to costs inversely proportional to the probabilities. The two costs are aggregated using a weighted product function. Let $w_1(ij)$ and $w_2(ij)$ be the costs produced by the first and second mapping systems respectively, the proposed aggregation function is given by equation 2:

$$w_{ij} = w_1\,(ij)^{\tau}\ w_2(ij) \qquad (2)$$

Where w_{ij} is a total link cost between node i and node j. Cost 1 is $w_1(ij)$, Cost 2 is $w_2(ij)$ while τ is the normalized weight. FS1 takes the normalized residual energy RE_i of node i and produces output variable relay probability_1 ($RP_1\ (ij)$) FS2 takes two inputs including the transmission energy TE(ij) consumed by node i to transmit to node j and the energy drain rate DR(i) of node i. It produces output variable relay probability_2 ($RP_2\ (ij)$) as shown in equation 3 and 4:

$$w_1(ij) = {}^{1}/_{RP_1\ (ij)} \qquad (3)$$

$$w_2(ij) = {}^{1}/_{RP_2\ (ij)} \qquad (4)$$

Since cost describing residual energy $w_1(ij)$, is more critical and hence τ is set to more than 1.
The approach DEFL greatly reduces the communication cost and improves the network lifetime of wireless sensor networks. The simulation results demonstrate the performance improvements obtained by DEFL when compared to other conventional algorithms such as Flow Augmentation (FA) algorithm [2], minimum total energy (MTE) routing approach [3][4] and minimum drain rate (MDR) mechanism [5].

2.2 A Mine WSN Low Consumption Acquisition Method Based on Neural Network

In this work [12], the author proposed a new wireless sensor node work mode to reduce the power consumption of the wireless sensor nodes in the mine monitoring system, a multi-model acquisition method based on neural network is proposed.
the power P at the transmitting terminal of the wireless sensor network can be constructed as the following function at equation 5:

$$P = f(R, \sigma) \qquad (5)$$

Where (R) is RSSI (Received Signal Strength Indicator) value and (σ) is the Standard deviation of the data.
The neural network can approximate such relationship. The optimization principle of neural network is to construct an adaptive linear neural network with an input layer of two nodes (RSSI value and σ) and an output layer of one node (power P). To train the data acquired in neural network, if the generalized result is consistent and the output power consumption is reduced.

2.3 Lightweight Reinforcement Learning for Energy Efficient Communications in Wireless Sensor Networks

In this work [10], High-density communications in wireless sensor networks (WSNs) demand for new approaches to meet stringent energy.
The authors proposed Q-Learning MAC (QL-MAC) protocol with the aim to extend the network lifetime; it is derived from Q-learning, which iteratively tweaks the MAC parameters

through a trial-and-error process to converge to a low energy state. This is achieved by adjusting the radio sleeping and active periods based on traffic predictions and transmission state of neighboring nodes.
The main beneficial effect of such Q-Learning based algorithm is adaptive duty-cycle that reduces the energy consumption over the time without affecting the other network performances, as shown by the simulations results.
Each node can independently determine an efficient wake-up schedule to limit as much as possible the number of slots in which the radio is turned on, thus greatly increasing its own (and overall network) lifetime.

2.4 Multipath routing through the firefly algorithm and fuzzy logic in wireless sensor networks

Multipath routing methods reduce energy consumption, improve scalability and provide load balancing in WSNs.
In this study [13], the authors suggested a multipath routing method for homogeneous WSNs. The proposed method includes3 phases: clustering the network nodes, discovering the paths between CHs, and maintaining the paths. In the first phase, wireless sensor network is clustered through the firefly algorithm. In the second phase, routing is performed between CHs based on the fuzzy logic. Routing between CHs results in creating 2 paths: primary path and backup path. CHs transmit data packets to the base station through the primary paths; however, failures in primary paths cause CHs to employ backup paths. In the third phase, the paths are maintained so that path breakages cause to restart route discovery. The results of the simulation reveal that the proposed multipath routing outperforms other routing methods in end-to-end delay, energy consumption, packet loss rate, and network lifetime.

2.5 Multipath Adaptive Periodic Threshold-sensitive Energy Efficient Protocol for Wireless Sensor Networks

In this work [14], the authors proposed a Multipath Adaptive Periodic Threshold-sensitive Energy Efficient Network Protocol (MAPTEEN). This routing protocol allows multiple routes for the information to reach a certain destination. The multipath approach increases the reliability of the network, decreases the power consumption and increases the life time of the network.

The sensor network is organized into clusters and the routing algorithm allows a single hop from one cluster to another all the way to the Base Stations (BS).

The communication between the nodes and the BS is carried out using the IEEE 802.15.4 protocol and the 2.4 GHz band (ISM band). The IEEE WPAN 802.15.4 (Wireless Personal Area Networks) is the media access control protocol layer used in our simulation model. This protocol is used for sending data between all the nodes and the base station.
The MAPTEEN protocol has 3 different packet types:

- Setup packet: sent by the BS in the setup phase. It contains the BS address.
- Setup reply packet: sent by nodes that received setup packet. It contains the node address and coordinates.
- Data packet: contains the source node address and the data (to the BS).

The performance of the MAPTEEN protocol: -

- The effect of the routing algorithm on the performance of the system is positive as seen by the less amount of the power used by the sensor nodes.
- The amount of data received by the base station increases linearly after the setup up phase.
- The life time of the nodes has improved over the other algorithms as our routing protocol is simple and uses relatively smaller distance between the nodes.
- The MAPTEEN algorithm has high reliability as it utilizes multiple paths to the next node.

2.6 Back propagation neural network (BPNN) based reliable routing method

In this work [15], the authors present reliable routing technique to find shortest path for data transmission using parameters like neighbor node distance and its energy. Back Propagation Neural Network algorithm is used to select the best nodes in path of data transmission. The algorithm is based on measuring the distance between source and destination and the residual energy consumption and selects the path with the lowest energy.

The route for the shortest path is computed for data transmission using the Euclidian distance (D) and the following equation is used to compute forwarding node's Energy conditions are shown in equation 6:

$$E_r(t) = E_i(t) - (0.1 \times D) \qquad (6)$$

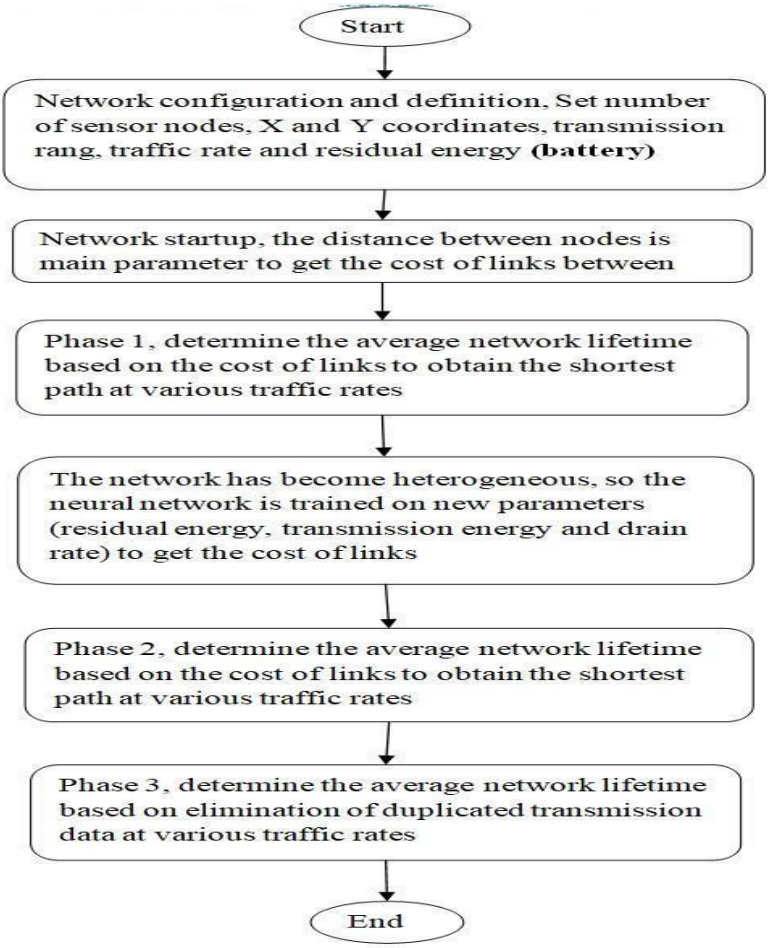
Where $E_r(t)$ is node's residual energy and $E_i(t)$ is node's initial energy

The energy consumption performance of neural data driven and the method of Back propagation neural network (BPNN) is as shown in the next Table. BPNN works by tuning the input/output parameters related to the distance and the remaining energy. The energy consumed is reduced by 27.2% on an average when compared to simple neural data driven method and hence the BPNN based method improves the performance in terms of energy consumption.

3 The Proposed Routing Algorithm

In this section, we describe the proposed routing algorithm based on a supervised machine learning module using neural networks that improves the lifetime of wireless sensor networks with heterogeneous nodes and variable traffic rates. This algorithm is based on shortest path routing with minimum cost links between nodes to reduce the effect of transmission energy and the drain rate. The transmission energy is reduced by selecting the route that utilizes the lower power consumption.

The next flowchart illustrates the algorithm:-



In the beginning and when operating the wireless sensor network, all nodes are homogeneous in the battery energy level, so the different distances between the nodes are the main factor in influencing the data transmission energy that consumes battery energy for each node, so the cost of the link between each node is determined according to the distance between one node and the other.

Equation 1 determines the relationship between distance and transmission energy $E_t(ij)$between node i and node j.

Based on the equation 1, we execute the initial phase of our proposed routing algorithm, which we refer to as Phase 1. In this phase we determine the average lifetime of the network's shortest paths based on the cost of the links between the nodes according to the distance in between sensor nodes.

Phase 2 begins as the network becomes heterogeneous sensor nodes in the battery energy level, so the neural networks are trained based on residual energy in node i, transmission energy between node i and node j, and drain rate of node i to get the cost of links between nodes

In determining the link cost from node i to node j, the following input variables are used:

1. The residual energy RE_i, It indicates the energy level of node i.
2. The transmission energy TE_{ij}, It represents the energy needed to transmit a data unit from node i to node j.
3. The energy drain rate DR_i, It indicates the rate of energy consumption of node i based on its traffic conditions.

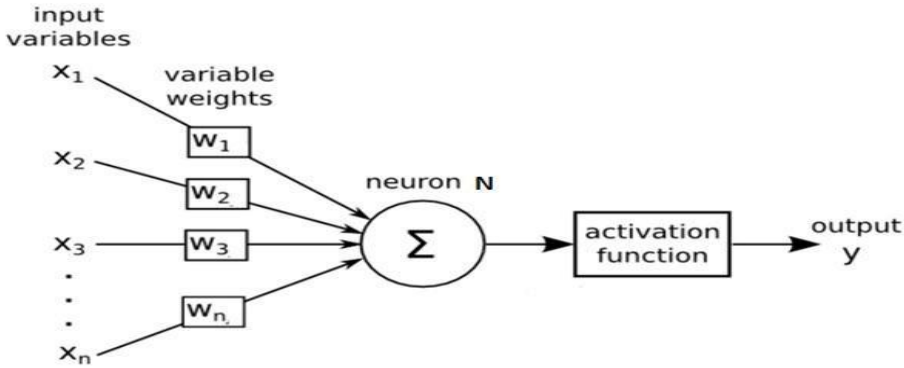


Fig. 2. Neural Perceptron Diagram

$$y = f\left(\sum_{m=1}^{n} x_m \, w_m^{(number\ of\ layer)}\right) \tag{7}$$

Where $x_m$inputs, m is a number of inputs, w are weights in equation 7 and f is the activation function is Logistic (sigmoid) function in equation 8:

$$f(t) = \frac{1}{(1 + e^{-t})} \tag{8}$$

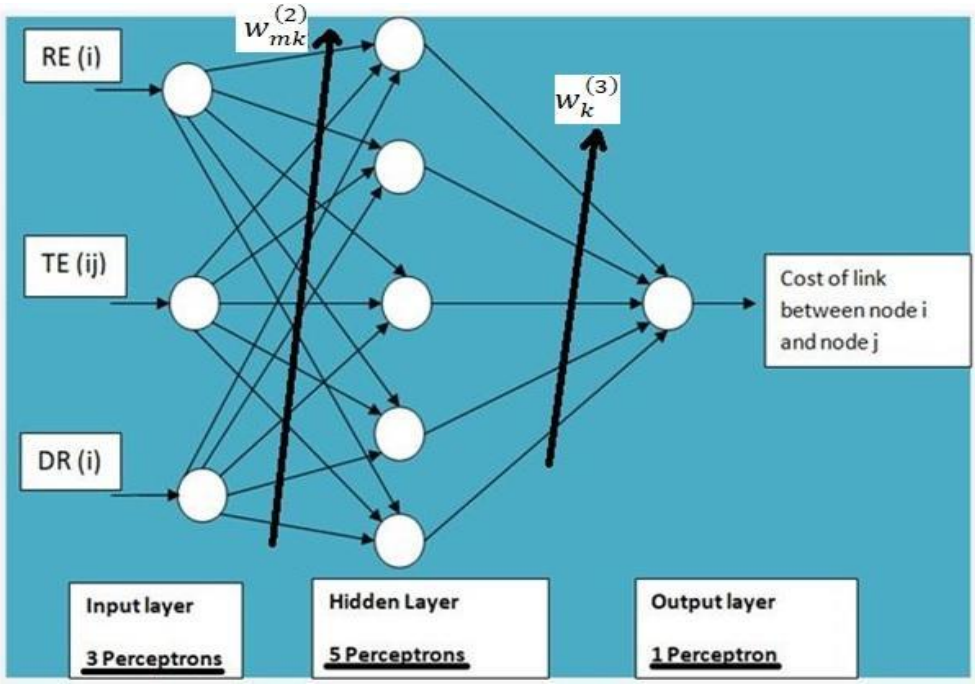


Fig. 3. Neural Network Diagram

$$a_k^{(2)} = f\left(\sum_{m=1}^{n} x_m w_{mk}^{(2)}\right) \quad (9)$$

$$h = f\left(\sum_{k=1}^{k} a_k^{(2)} w_k^{(3)}\right) \quad (10)$$

where k is the number of perceptrons of the hidden layer, $a_k^{(2)}$ is the output of hidden layer, $w_{mk}^{(2)}$ is weights of hidden layer in equation 9, $w_k^{(3)}$ is weights of output layer and h is the cost of the link between node i and node j which is the output of neural network in equation 10 .

Phase 3 begins for more enhancement of a network lifetime by reducing transmission energy via eliminating the duplicated transmission data between sensor nodes, using XOR logic circuit embedded in each node.

4 Results and Simulations

In this section, the performance of our proposed algorithm is evaluated in terms of average network lifetime and average energy consumption rate, for different traffic rates.

We implemented the proposed routing algorithm using Google Colab simulator connected to python 3 Google compute engine backend.

A summary of the simulation parameters is described below in Table 1. Note that these parameters are comparable to the parameters used in the reference algorithms.

TABLE I. SIMULATION PARAMETERS

Parameter	value
Topological area	40m X 40m
Number of nodes	10 sensor nodes
Transmission range	20 m
Data packet size	500 bits
Energy consumed at the transmitter amplifier	100 PJ/bit/m4
Energy dissipated in transmitting circuitry	50 nJ/bit
Residual Battery Energy	8 J
Traffic Rate	5, 10, 20, 30, 40, 50 Packets/Sec
X and Y coordinators from node 0 to node 9	**(33,20) - (32,35)- (40,20) - (32,5) - (25,30) - (25,10) - (15,40) - (10,35) - (10,10) - (15,0)**
Data duplicated ratio	**10%, 20%, 30%**
Microcontroller used in node chip	Atmel AVR 8-bit Atmega32 Microcontroller

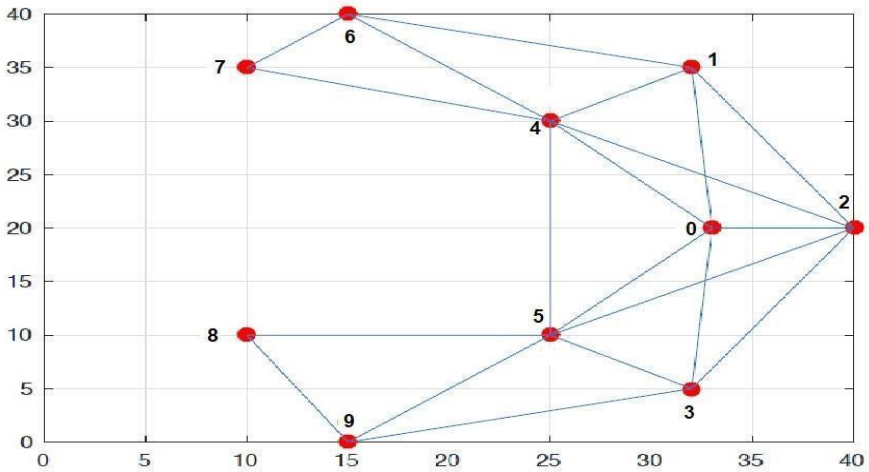


Fig. 4. A Wireless Sensor Network Area

In Fig. 4, we illustrate area deployment of the WSN nodes. In Fig. 5, we further show the lifetime performance for different traffic rates

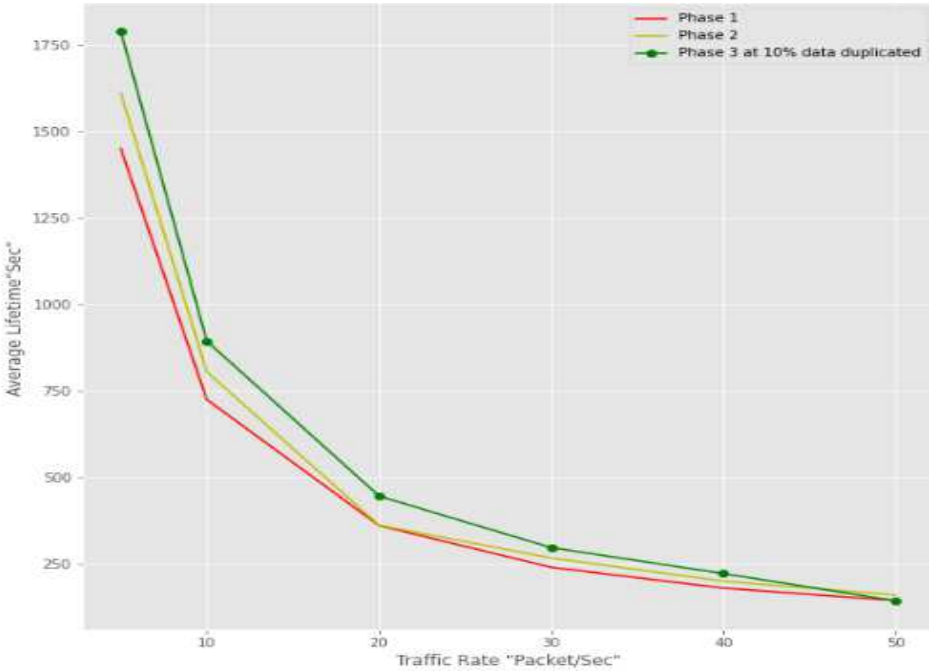


Fig. 5. Average network lifetime performance of different traffic rates

In Fig. 6, we further show the lifetime performance for different traffic rates and data duplicated percentages.

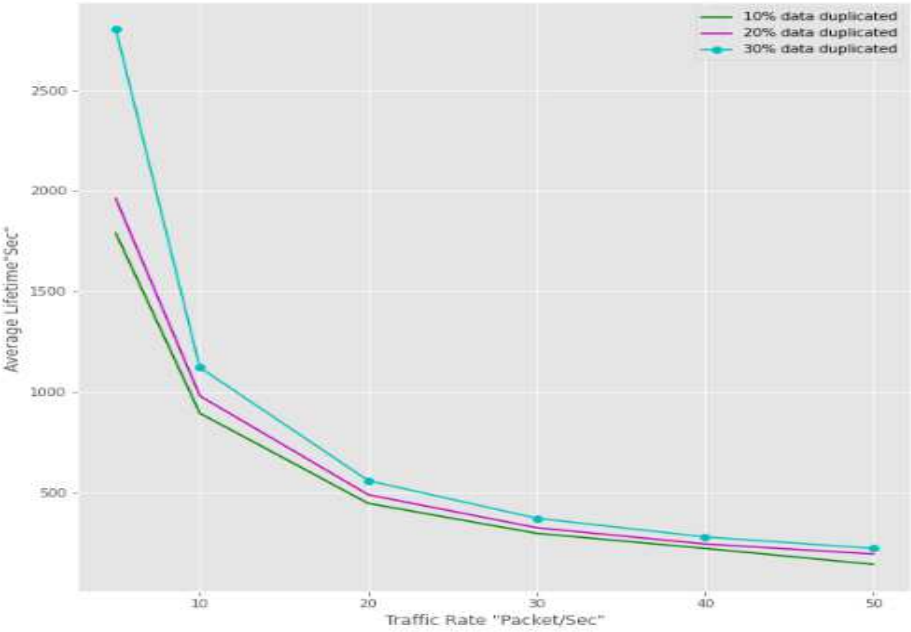


Fig. 6. Average network lifetime performance of different data duplicated and traffic rates

In Fig. 7 show that our proposed algorithm performance is better than DEFL was proposed in [11], the minimum drain rate (MDR) mechanism was proposed in [16] and Flow Augmentation (FA) algorithm was proposed in [17].

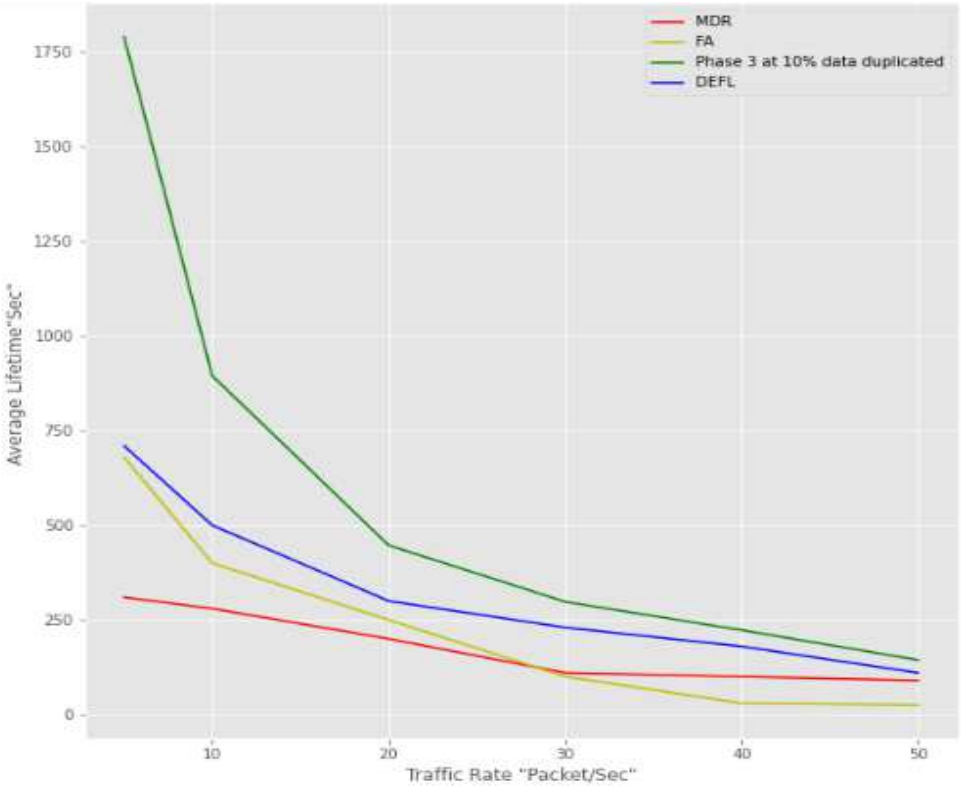


Fig. 7. Comparison between proposed algorithm phase 3 at 10 % data duplication and DEFL, FA and MDR algorithms performance of different traffic rates

This paper assumed that the power consumption in computing processing of algorithm is neglected according to use Atmega32 microcontroller that consumed 1.1mA at 1MHz, 3V and 25°C [18].

5 Conclusion

This paper presented a supervised neural network machine learning routing algorithm that get the shortest path between sensor nodes of network and added more enhancement to the network lifetime by reducing transmission energy via eliminating the duplicated transmission data between sensor nodes. The newly developed algorithm increases the lifetime of the network by up to 75%, depending on the traffic rate, over the DEFL algorithm.

The experimental results show that the new neural network supervised machine learning routing algorithm that get the shortest path between sensor nodes of network improve the lifetime of the network by more than 68% over existing algorithm. The experimental results also show that after using the elimination factor of the duplicated data. Our algorithm improves the lifetime of the network and improves the performance over the DEFL, FA, and MDR algorithms.

Declarations

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Conflicts of interest

The authors declare that they have no competing interests.

Availability of data and material

The datasets collected and/or analyzed during the current study are available from the corresponding author on request.

Code availability

The custom code used in this research is available from the corresponding author on request.

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